Serverless und Event-Driven Architecture

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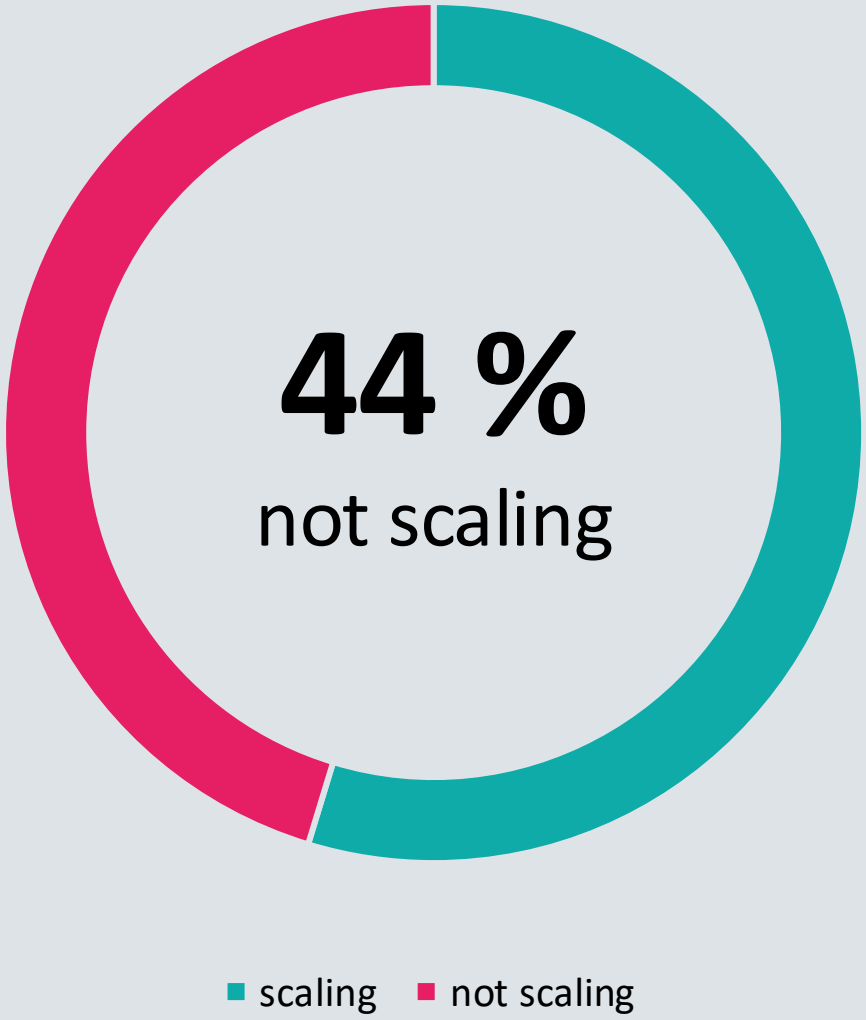
Pets vs. Cattle

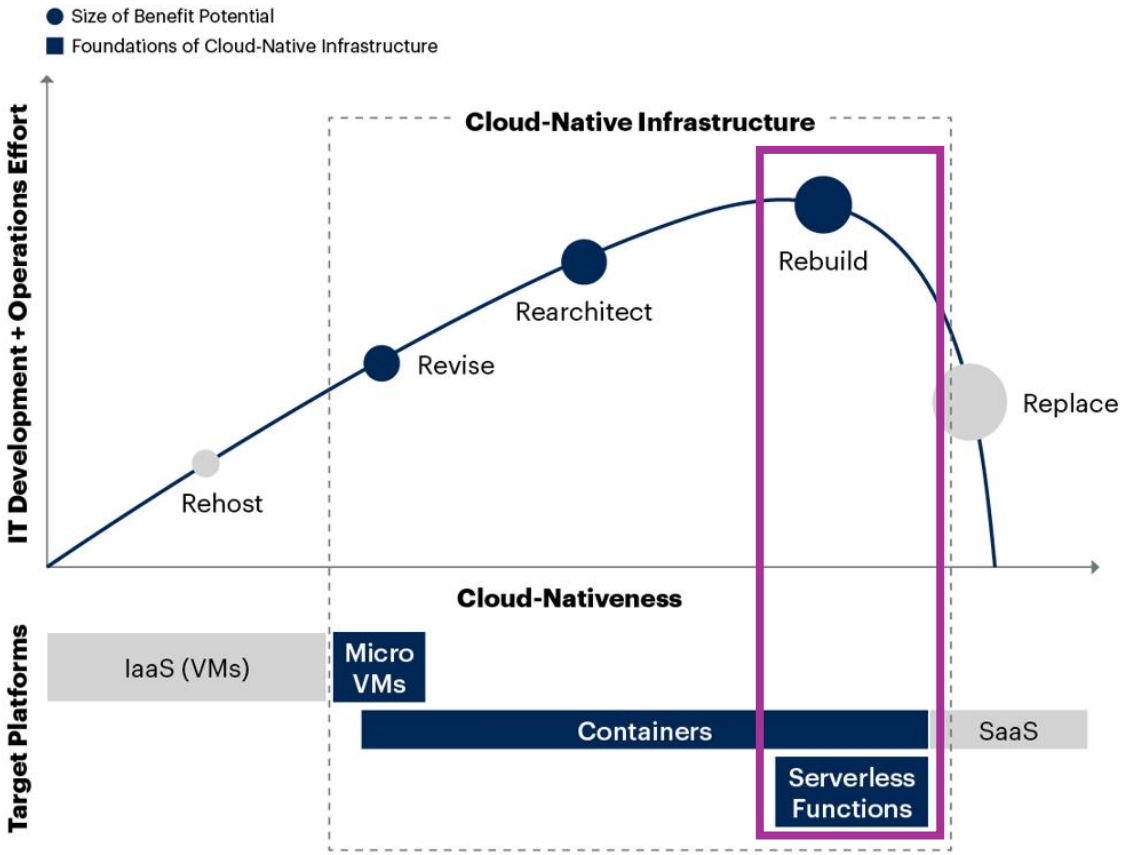


Pets or Cattle?

44% of the AKS Node Pools for a major Azure client are configured to operate without auto-scaling, meaning they maintain a fixed number of virtual machines running 24/7.

Serverless means rebuild

Virtualization Abstractions for Cloud-Native Infrastructure
Illustrative



Source: Gartner
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Gartner

Serverless is more than just FaaS

Serverless is an **entire stack of services** that can respond to specific events or requests, and **scale to zero** when no longer in use—and for which **provisioning, management and billing are handled by the cloud provider** and invisible to developers.

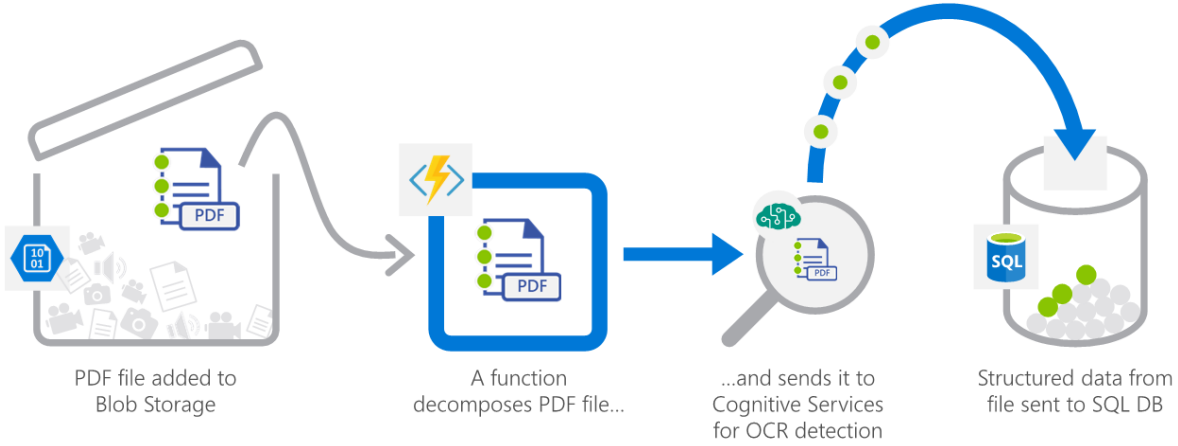
Function-as-a-service, or FaaS, is a cloud computing service that enables developers to run code or containers in response to specific events or requests, without specifying or managing the infrastructure required to run to code.

FaaS is the compute model central to serverless, and the two terms are often used interchangeably. **But serverless is much more than FaaS.**

Source: https://www.ibm.com/topics/serverless

Azure Functions
Azure Logic Apps
Azure App Service (with Consumption Plan)
Azure Cosmos DB
Azure Event Grid
Azure SQL Database Serverless
Azure SignalR Service
Azure Cognitive Services (some features can be used in a serverless manner)
Azure Key Vault
Azure Search
Azure API Management (Consumption Tier)
Azure Notification Hubs
Azure IoT Hub
Azure Machine Learning Service (in certain serverless configurations)
Azure Container Instances (ACI)
Azure Durable Functions
Azure Stream Analytics
Azure Cognitive Search
Azure Data Lake Analytics
Azure Data Factory (V2)
Azure DevOps (Azure Pipelines)
Azure Service Bus (Auto-Inflate feature for scaling)
Azure WebJobs
Azure App Configuration
Azure Static Web Apps
Azure API Apps
…

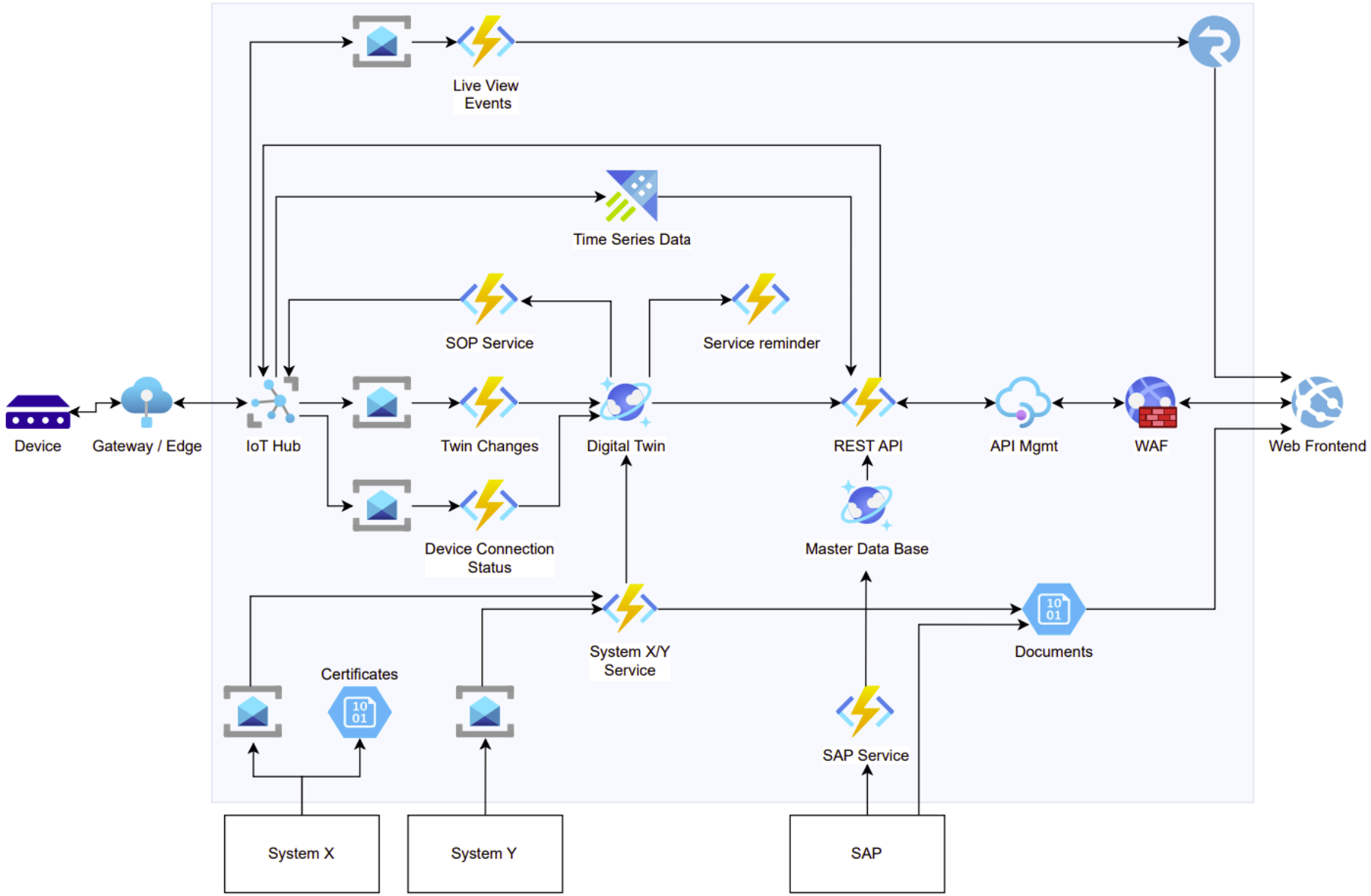
Serverless scenario: File upload and document processing



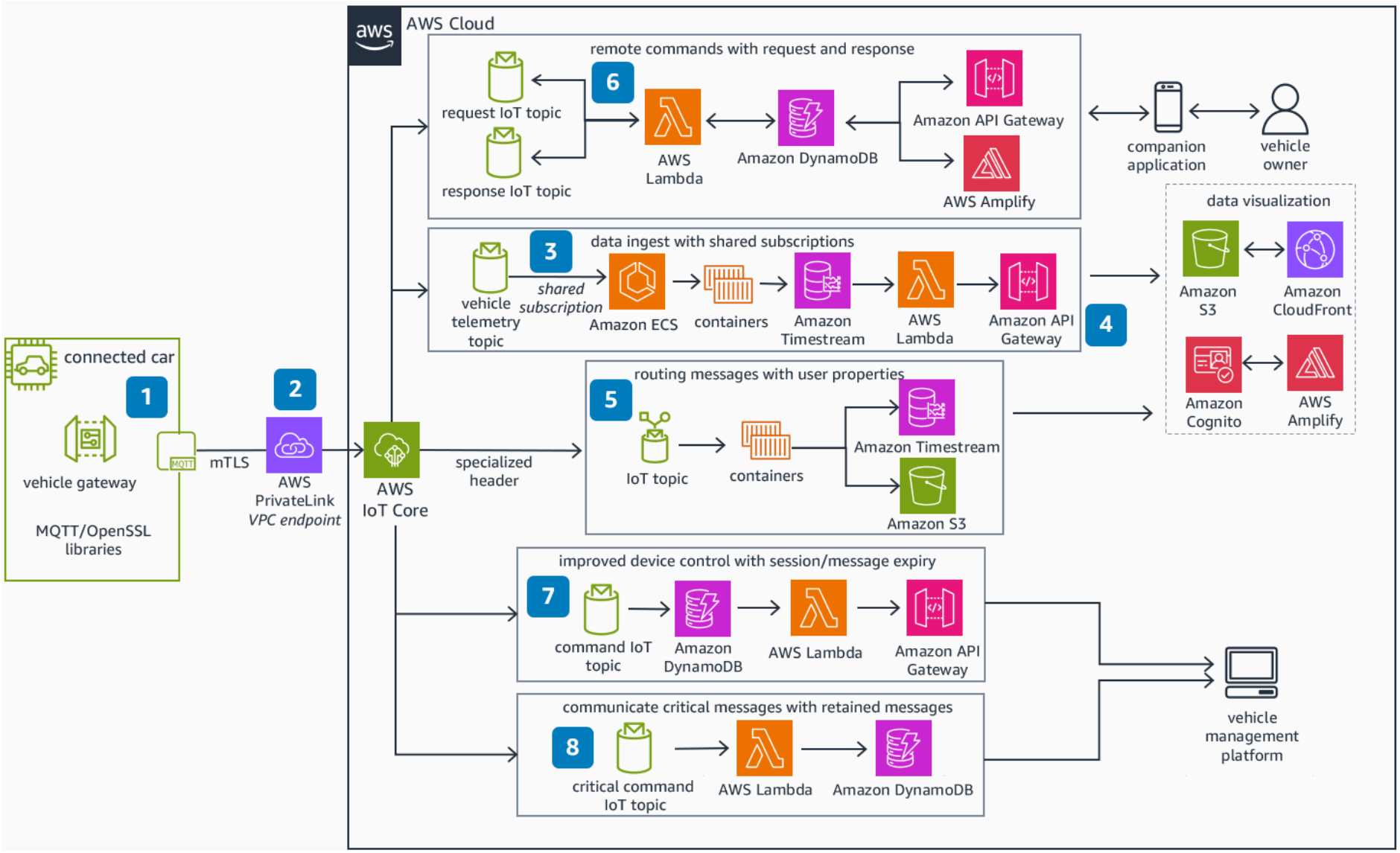
Real-time file processing for the healthcare scenario: The solution securely uploads patient records as PDF files. The solution then decomposes the data, by processes it using OCR detection, and it adds the data to a database for easy queries.

Source: https://learn.microsoft.com/en-us/azure/architecture/serverless-quest/reference-architectures

Serverless scenario: IoT

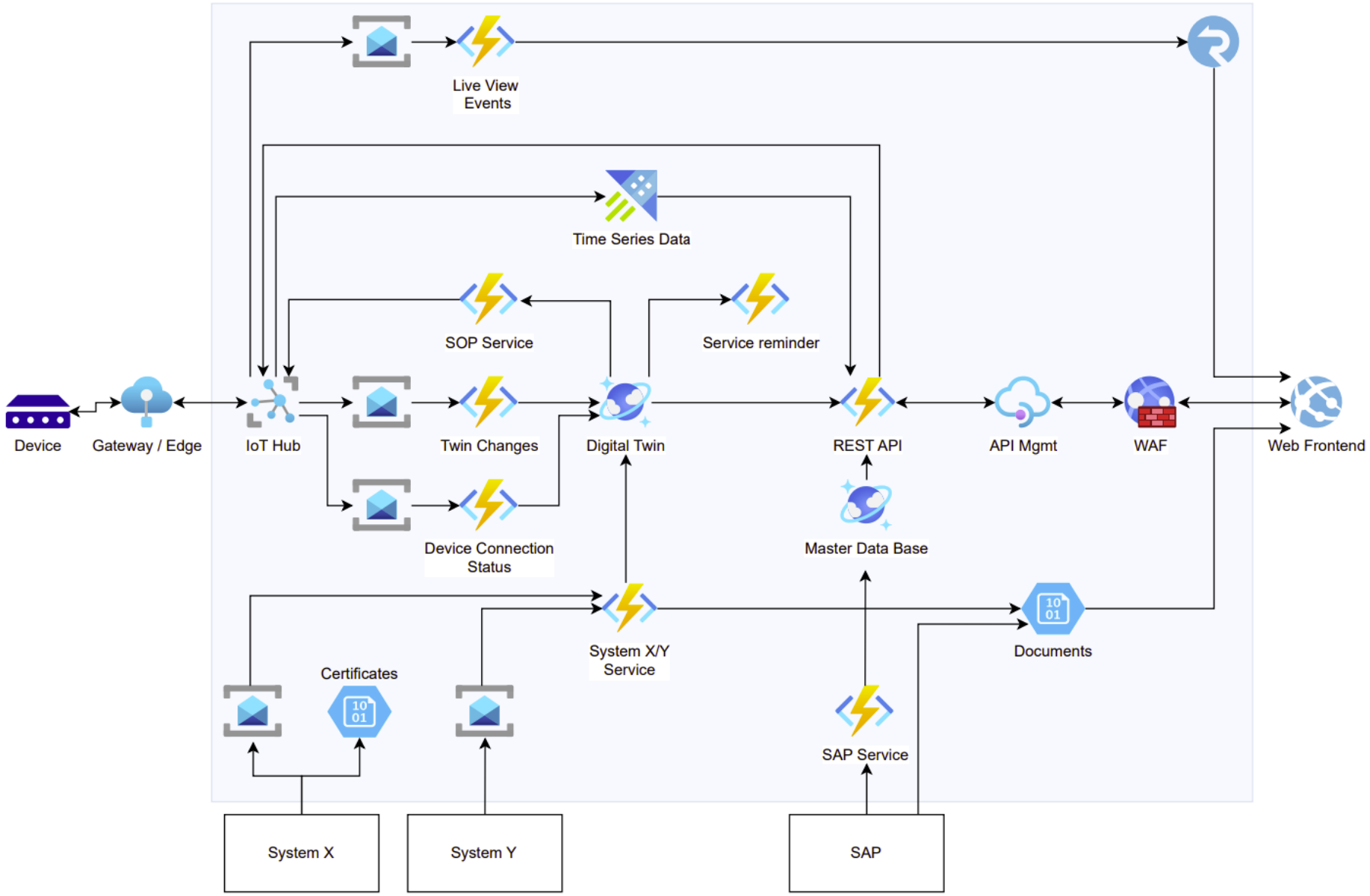


Serverless scenario: AWS Connected Vehicle – Reference Architecture



Source: https://docs.aws.amazon.com/architecture-diagrams/latest/aws-connected-vehicle/aws-connected-vehicle.html

Serverless scenario: IoT



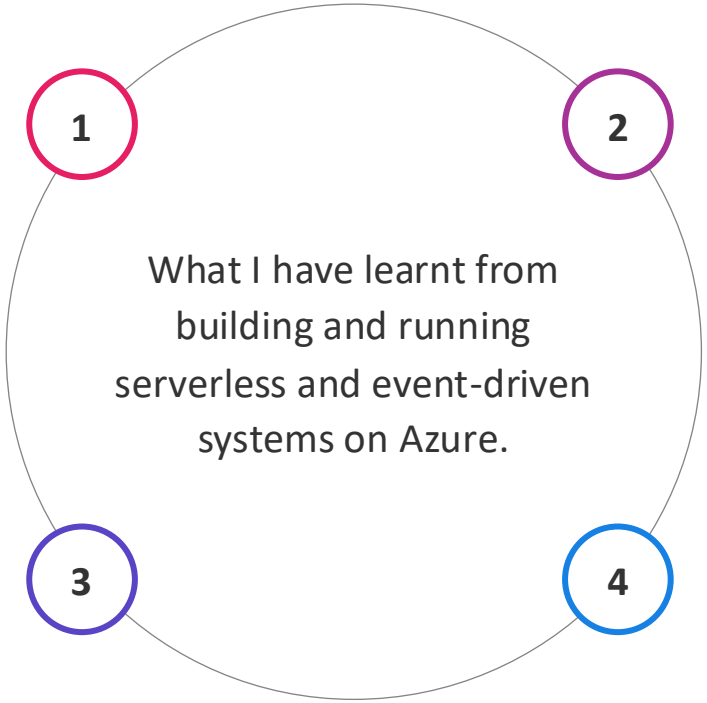
Challenges and solution approaches

Security

Adding Infrastructure to re-establish Perimeter-Security vs. implementing SaaS-style Zero-Trust.

Cloud Automation

For serverless, cloud automation (aka. IaC) needs to focus on the application, the relationships between component and the configuration instead of the infrastructure.



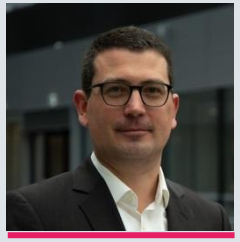
Operations

Complexity has moved from code to relationships between the components. Monitoring serverless and event-driven systems end-to-end is complex.

Architecture

Reconsider traditional Software Design Patterns for serverless architecture, e.g. Messaging (Pipes-and-Filters), Hexagonal Architecture and CQRS.

Questions and Answers



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